NAME ____________________ DATE ____________________ PERIOD ____________________

ISS Greenhouse Module

Campaign 1 · Case 01 · Low Earth Orbit

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Accessible Mission

MISSION QUESTION
Why are the lettuce roots tangled and directionless even though water, nutrients, temperature, and lighting appear normal?

SCIENCE FOCUS
Plants normally use gravity to help guide root and shoot growth. In microgravity, that directional cue becomes unreliable, so engineers must provide another way for plants to orient.

REFERENCE

1 · Vocabulary

Gravitropism	Directional growth in response to gravity.
Microgravity	Apparent weightlessness during continuous free fall in orbit.
Statocyte	Plant cell involved in sensing gravity.
Statolith	Dense starch-filled structure that helps provide a direction cue.
Phototropism	Directional growth in response to light.

PREDICTION

2 · Initial thinking

What evidence would help you tell a resource problem from an orientation problem?

You may answer with words, phrases, or bullet points.

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EVIDENCE TASK

3 · Investigate four evidence sources

Record only evidence that helps support or reject a cause. You may use phrases or bullet points.

Source	Observation or data	What does it support or reject?
Crew		
Sensors		
Plants		
Mission logs		

NOTES

CAUSE TEST

4 · Test the competing explanations

For each explanation, mark supported or weakened. Then give one reason.

Nutrient solution is damaging roots.

Status and one evidence-based reason.

The light cycle is wrong.

Status and one evidence-based reason.

The seed stock is defective.

Status and one evidence-based reason.

Microgravity disrupts orientation.

Status and one evidence-based reason.

MECHANISM MODEL

5 · Build the mechanism

WORD BANK curve or grow without consistent orientation · downward · settle · settle in one direction

Earth gravity

Statoliths ______

Roots grow ______

Microgravity

Statoliths do not ______ consistently.

Roots ______

DIAGNOSIS

6 · Diagnose and reject an alternative

What caused the tangled, directionless roots?

Use the environmental condition and the plant mechanism.

ALTERNATIVE

6 · Diagnose and reject an alternative

Which other explanation is tempting, and what evidence weakens it?

You may answer in two sentences or bullet points.

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EXPLANATION

7 · Claim-Evidence-Reasoning

You may write sentences or use bullet points. Use evidence from more than one source.

CLAIM	
EVIDENCE	
REASONING	

ENGINEERING RESPONSE

8 · Supply a consistent orientation cue

Choose or design a system that gives roots a consistent cue.

Describe or draw your design.

Examples include root channels, a moisture gradient, a root guide plus lighting, or a rotating chamber.

One criterion

What must the design accomplish?

One constraint

What limit must the design work within?

TRANSFER CHECK

9 · Exit ticket

Engineers add narrow root channels but do not change the lighting. Which symptom should improve first: tangled roots or pale leaves? Explain.

OPTIONAL EXTENSION

Compare a physical root guide with a rotating growth chamber. Which is easier to use on the ISS, and which more closely replaces gravity?